



main.c

Output



```
1 // Elsa Mariya George, Statistics
2 #include <stdio.h>
3
4 int main() {
5     int a[100][100],m,n,i,j;
6
7     printf("enter row number and column number.");
8     scanf("%d%d",&m,&n);
9
10    printf("Enter matrix elements: ");
11
12
13    for(i=0;i<m;i++)
14    {
15        for(j=0;j<n;j++)
16        {
17            scanf("%d",&a[i][j]);
18        }
19    }
20    for(i=0;i<m;i++)
21    {
22        for(j=0;j<n;j++)
23        {
24            printf("%d\t",a[i][j]);
25        }
26        printf("\n");
27    }
28    printf("the diagonal elements are: ");
29
30    for(i=0;i<m;i++)
31    {
32        for(j=0;j<n;j++)
33        {
34            if(i==j)
35            {
```

Run



main.c

Output



```
12
13     for(i=0;i<m;i++)
14     {
15         for(j=0;j<n;j++)
16         {
17             scanf("%d",&a[i][j]);
18         }
19     }
20     for(i=0;i<m;i++)
21     {
22         for(j=0;j<n;j++)
23         {
24             printf("%d\t",a[i][j]);
25         }
26         printf("\n");
27     }
28     printf("the diagonal elements are: ");
29
30     for(i=0;i<m;i++)
31     {
32         for(j=0;j<n;j++)
33         {
34             if(i==j)
35             {
36                 printf("%d\t",a[i][j]);
37                 break;
38             }
39         }
40         printf("\n");
41
42
43
44     }
45     return 0;
46 }
```

Run



main.c

Output



```
enter row number and column number.3
```

```
3
```

```
Enter matrix elements: 4
```

```
5
```

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6
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4 5 6
```

```
7 8 9
```

```
2 4 5
```

```
the diagonal elements are: 4
```

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8
```

```
5
```

```
=== Code Execution Successful ===
```



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English (India)

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